Label-Efficient Illicit Transaction Detection on the Elliptic Bitcoin Graph via Self-Supervised Pretraining

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Abstract*—Anti-money laundering (AML) detection in blockchain transaction networks is severely hindered by extreme class imbalance, severe label scarcity, and temporal distribution shifts. Using the Elliptic Bitcoin benchmark, this study investigates how Self-Supervised Learning (SSL) on graph structures mitigates these challenges. We pretrain a Graph Convolutional Network (GCN) via a node-level contrastive objective on the full graph topology without relying on ground-truth labels. To address domain-specific constraints, we incorporate two methodological extensions: (i) Extension Track B1 (Knowledge Distillation), transferring dark knowledge from a fully supervised teacher to regularize representations under extreme label scarcity (10% labels), and (ii) Extension Track A2 (Domain-Adversarial Training), employing a gradient reversal layer to enforce temporal invariance across the dark-market shutdown at time step 43. Experimental evaluations show that SSL combined with Knowledge Distillation (B1) increases the illicit F1-score from 0.4509 (supervised baseline) to 0.4763 under 10% label supervision. Furthermore, standalone SSL pretraining substantially reduces temporal drift, retaining nearly double (2.1×) the post-shutdown illicit F1-score compared to the supervised baseline, while Domain-Adversarial Training (A2) yields a more modest robustness gain (1.3×).

Keywords*—Self-Supervised Learning, Graph Neural Networks, Anti-Money Laundering, Elliptic Dataset, Domain-Adversarial Training, Knowledge Distillation.

I. INTRODUCTION

Financial forensics and Anti-Money Laundering (AML) on public blockchains present fundamental graph anomaly detection challenges [1]. The Elliptic Bitcoin dataset provides a temporal transaction graph of 203,769 nodes and 234,355 directed edges across 49 distinct time intervals [1]. Each node represents a transaction with 166 handcrafted features [1]. Despite the rich structural context, standard supervised paradigms face two major bottlenecks:

1) Label Scarcity: Only 2% of transactions are confirmed illicit, while 77% remain completely unlabeled [1]. Supervised classifiers discard this extensive structural connectivity during training [1].

2) Temporal Distribution Shift: At time step 43, a major dark-market shutdown altered the behavioral patterns of illicit entities, leading to severe concept drift and catastrophic performance degradation for supervised models deployed on subsequent steps [1].

To resolve these limitations, this paper reformulates illicit transaction detection through Graph Self-Supervised Learning (SSL) [1]. We pretrain a shared encoder on the full topological structure and evaluate its transferability under varying supervision fractions and temporal splits [1]. Furthermore, we implement two targeted extensions from the project scope: Extension Track B1 (Teacher-Student Knowledge Distillation) to maximize low-budget label efficiency, and Extension Track A2 (Domain-Adversarial Learning) to induce temporal domain invariance. The source code and execution pipeline are publicly available at: https://github.com/amiirabbasdashtbozorg/elliptic-ssl-anomaly-detection

II. RELATED WORKS

Supervised graph learning on the Elliptic dataset was pioneered by Weber et al. [1], establishing baseline benchmarks using Random Forests and Graph Convolutional Networks (GCNs). Alarab et al. [2] enhanced supervised GCN representations by concatenating convolutional layer outputs with linearly projected input features.

In self-supervised graph representation learning, Deep Graph Infomax (DGI) [5] and GRACE [4] demonstrated that maximizing mutual information between perturbed graph views yields rich, transferable representations without manual annotation. This work adapts node-level contrastive learning [4] to financial networks, integrating adversarial domain adaptation and knowledge distillation.

Chen et al. [8] proposed MDST-GNN, a multi-distance spatial-temporal graph neural network with an embedded self-supervised objective, reporting AUC gains on Elliptic over prior methods but without a separate contrastive pretrain-then-fine-tune stage. Yang et al. [9] introduced Samen, a state-space framework that addresses temporal concept shift in dynamic graphs through prefix-suffix state learning, offering an architecture-level alternative to our gradient-reversal-based Extension A2.

III. METHODOLOGY

Our proposed architecture comprises a supervised reference model, a contrastive pretraining framework, and two specialized extensions.

A. Supervised Baseline

Following Kipf & Welling [3], we deploy a 2-layer GCN. The node propagation rule is defined as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right)$$

where $\tilde{A} = A + I_N$ is the adjacency matrix with added self-loops and $\tilde{D}$ is the degree matrix. Inverse class-frequency weighting is incorporated into the Cross-Entropy loss to counteract severe class imbalance.

B. Self-Supervised Pretraining (SSL)

To exploit the 77% unlabeled nodes, we pretrain a GCN encoder f_θ via GRACE-style contrastive learning [4]. At each epoch, two stochastic graph views $(\tilde{X}_1, \tilde{A}_1)$ and $(\tilde{X}_2, \tilde{A}_2)$ are generated via uniform edge dropping (p=0.2) and feature-dimension masking (p=0.1). Node embeddings are mapped via a 2-layer MLP projection head $g(\cdot)$ into a latent space. A symmetric InfoNCE (NT-Xent) loss maximizes agreement between positive pairs (the same node across views) while pushing apart distinct node embeddings in the batch:

$$L_{InfoNCE} = -\log \frac{\exp\left(\text{sim}\left(z_i, z_j\right)/\tau\right)}{\sum_k \exp(\text{sim}(z_i, z_k)/\tau))}$$

C. Extension Track B1: Teacher-Student Knowledge Distillation

To bolster performance under severe label scarcity, we introduce Extension B1 during downstream fine-tuning. The fully supervised baseline (trained on 100% labels) serves as a frozen Teacher model. The student model (the pretrained SSL encoder + classifier head) is optimized using a composite objective:

$$L_{Total} = L_C E + \alpha \cdot T^2 \cdot D_{KL}\left(\sigma(z_s/T) \parallel \sigma(z_t/T)\right)$$

where D_{KL} denotes Kullback-Leibler divergence, T=2.0 is the distillation temperature, and $\alpha = 0.5$ balances hard label supervision with soft dark-knowledge regularization.

D. Extension Track A2: Domain-Adversarial Regularization

To counteract temporal degradation across the time step 43 market shutdown, Extension A2 incorporates a Domain Discriminator connected to the encoder representations via a Gradient Reversal Layer (GRL). The discriminator optimizes a binary classification task to distinguish pre-shutdown ($t < 43$) from post-shutdown ($t \geq 43$) representations. The GRL inverts gradient updates ($\times -\lambda$) during backpropagation, compelling the encoder to discard time-specific artifacts.

IV. EXPERIMENTAL SETUP

The dataset is partitioned chronologically: time steps 1–34 form the training graph, while steps 35–49 constitute the test set [1]. The test split is further divided into pre-shutdown (steps 35–42) and post-shutdown (steps 43–49) subsets to measure robustness [1]. Evaluation focuses on the illicit class using Precision, Recall, and minority-class F1-score [1].

V. RESULTS AND DISCUSSION

A. Label-Efficiency Analysis

Table I and Fig. 1 summarize model performance across varying training label fractions (100%, 50%, 25%, and 10%).

TABLE I. MASTER RESULTS TABLE: ILLICIT F1-SCORE ACROSS REGIMES

Model	*100% Labels*	*50% Labels*	*25% Labels*	*10% Labels*	*Pre-SD*	*Post-SD*
Supervised Baseline	0.6046	0.5737	0.4952	0.4509	0.6840	0.0270
SSL (Fine-tuned)	0.5978	0.5701	0.5706	0.4211	0.6998	0.0567
SSL + A2 (Domain Adv)	0.5818	0.4757	0.5190	0.4320	0.6832	0.0349
SSL + B1 (Distillation)	0.6067	0.5552	0.5701	0.4763	0.6962	0.0228
SSL + A2 + B1	0.6019	0.5483	0.5518	0.4754	0.6861	0.0310

Under extreme label scarcity (10% fraction), the supervised baseline drops to an illicit F1 of 0.4509. While standalone SSL achieves 0.4211, the SSL + B1 (Distillation) framework establishes the highest performance at 0.4763. This confirms that soft teacher supervision effectively regularizes the latent space when annotations are scarce.

A plausible cause is a mismatch between the pretraining objective and the downstream task: contrastive learning captures dominant structural patterns rather than the subtler cues needed to separate illicit from licit nodes, and with only 2% of nodes labeled illicit, the classifier head has too little signal to reorient this generic representation toward the minority class. This likely explains why SSL improves temporal robustness, where general invariance suffices, but not label efficiency, where fine-grained class discrimination matters more.

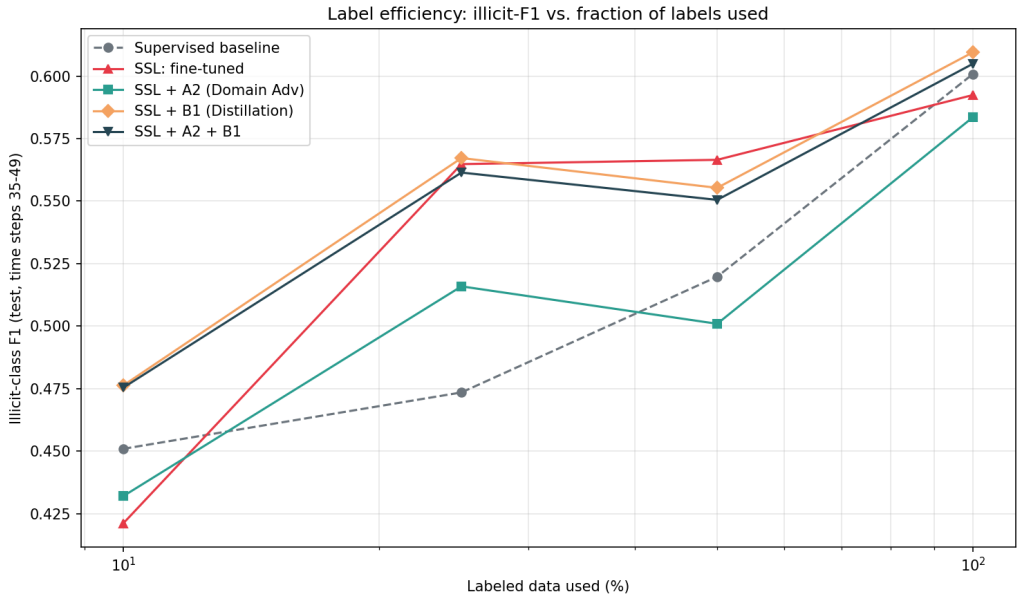


Fig. 1. Illicit-class F1-score across labeled data fractions (10% to 100%). SSL + B1 consistently outperforms the baseline under severe label constraints.

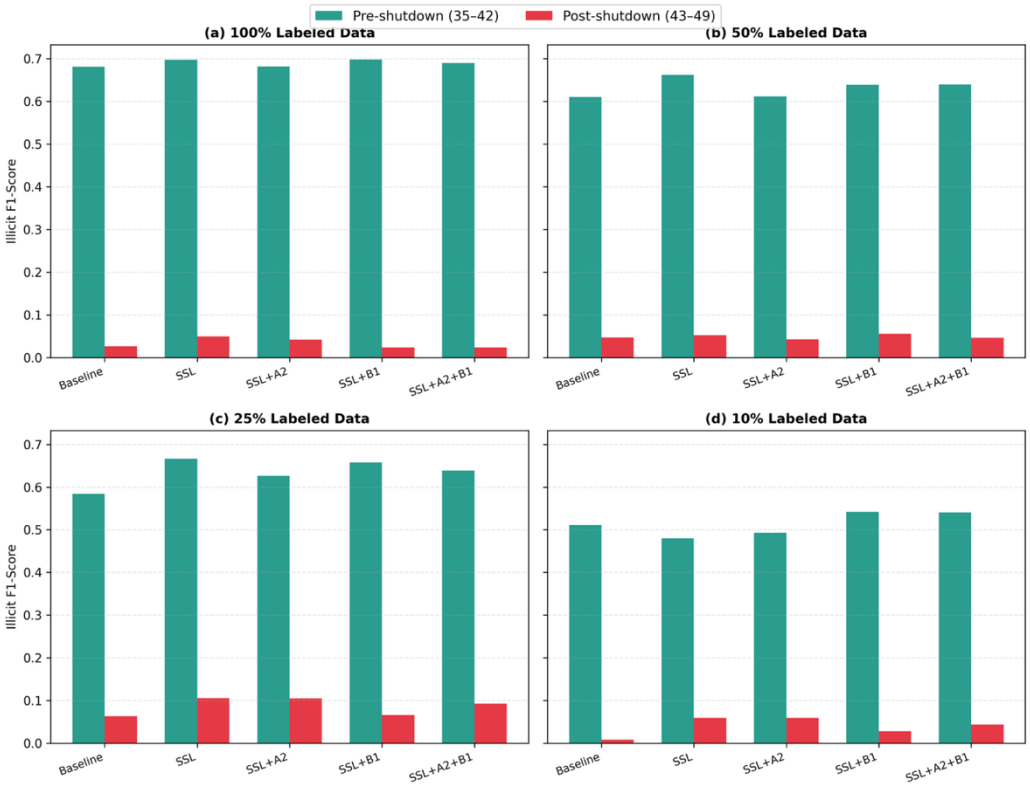


Fig. 2. Pre- vs. Post-shutdown illicit F1-score. SSL and SSL + A2 preserve substantially higher post-shutdown detection capabilities compared to the supervised baseline.

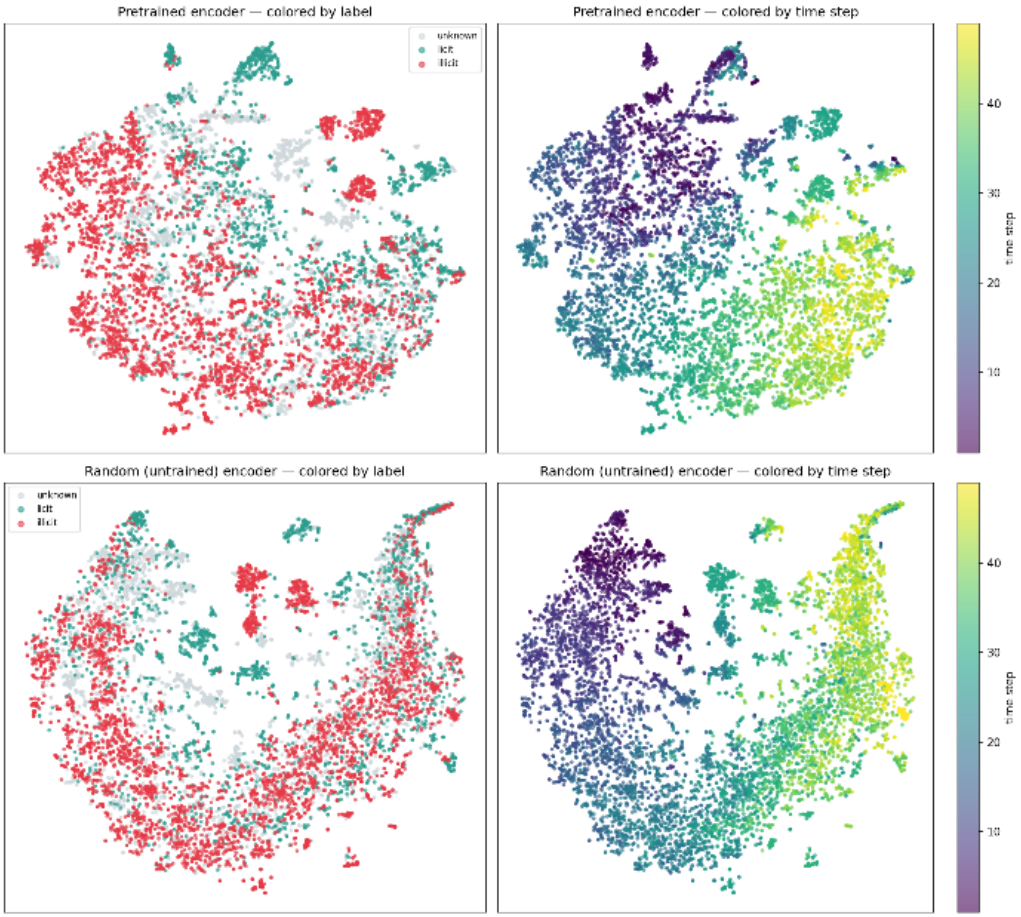


Fig. 3. t-SNE projection of pretrained node embeddings, colored by class (illicit / licit / unknown). Structurally similar transactions cluster together prior to any label exposure, indicating the SSL encoder captures topologically meaningful structure.

B. Temporal Robustness Evaluation

As detailed in Table I and Fig. 2, the baseline model experiences severe degradation across the shutdown boundary, plummeting by 96.1% from a pre-shutdown F1 of 0.6840 to 0.0270 post-shutdown.

In contrast, SSL (0.0567) and SSL + A2 (0.0349) retain approximately 2.1× and 1.3× the baseline's post-shutdown score. This demonstrates that pretraining on structural graph invariants shields the model from abrupt behavioral distribution shifts.

C. Latent Representation Visualization

t-SNE projections of the latent node representations demonstrate that the self-supervised contrastive objective clusters topologically coherent transaction structures prior to label exposure, validating the encoder's feature extraction quality.

D. Computational Overhead

Table II reports wall-clock training and inference cost, measured on a single NVIDIA GeForce MX150.

TABLE II. TRAINING AND INFERENCE TIME OVERHEAD

Stage	*Time/Epoch (ms)*	*Epochs*	*Total Time*
Supervised Baseline	508.3	200	101.7 s
SSL Pretrain (per scope)	5478.6	400	2191.4 s
SSL Pretrain (both scopes)	-	800	4382.9 s
SSL Linear Probe	216.5	200	43.3 s
SSL Fine-tune	486.4	200	97.3 s

SSL contrastive pretraining introduces a substantial one-time upfront cost: pretraining the encoder for 400 epochs over a single graph scope takes 2191.4 s (~36.5 minutes), rising to 4382.9 s (~73.0 minutes) when both the full-graph and train-range-only scopes are pretrained — roughly 43× the 101.7 s needed to train the supervised baseline from scratch. Critically, however, this cost is incurred only once: unlike the supervised baseline, which must be retrained from scratch at every label fraction, the pretrained encoder is reused directly across the linear probe, fine-tuning, and both extension experiments.

Downstream costs are comparable to, or cheaper than, the supervised baseline. Fine-tuning the pretrained encoder

(97.3 s) is essentially on par with training the baseline from scratch (101.7 s), while linear probing — which freezes the encoder and trains only a lightweight classification head — is more than 2× faster at 43.3 s. At inference time, the SSL fine-tuned model (193.2 ms per forward pass) adds only a 3.2% overhead over the baseline (187.1 ms). This confirms that the label-efficiency and temporal-robustness gains from SSL pretraining come at negligible deployment-time cost, with the entire overhead concentrated in the one-time pretraining stage.

Note that the combined-scope pretraining total approximates the train-only scope's cost as equal to the full-graph scope's; since the train-only graph has fewer nodes, its true cost is likely marginally lower than the 4382.9 s reported here.

VI. CONCLUSION AND FUTURE WORK

This study validates that self-supervised graph pretraining extracts essential topological representations from unlabeled blockchain transactions [1]. Knowledge Distillation (Extension B1) significantly enhances sample efficiency under label scarcity, while Domain-Adversarial Training (Extension A2) stabilizes predictions against temporal shocks [1]. Future directions include incorporating dynamic temporal GNNs (e.g., EvolveGCN [7]) and generative structure reconstruction objectives (DOMINANT [6]) [1].

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